EVALUATION

I Multiple Choice Questions

1. During aerobic cellular respiration, most of the ATP made is generated by

   a. Substrate level phosphorylation
   b. Pyruvate kinase
   c. Glycolysis
   d. Oxidative phosphorylation

2. The prosthetic group of NADH dehydrogenase is

   a. FMN
   b. NADH
   c. FAD
   d. NADPH

3. The chemiosmotic theory involves all except

   a. A membrane impermeable to protons
   b. Electron transport by the respiratory chain pumps protons out of mitochondria
   c. Proton flow into the mitochondria depends on ADP + Pi
   d. Only proton transport is strictly regulated. other positively charged ions can easily diffuse freely across the mitochondrial membrane

4. Which of the following accepts only one electron?

   a. Cytochrome b
   b. Coenzyme Q
   c. FMN
   d. FAD

5. Loss of electrons can be termed as

   a. Metabolism
   b. Anabolism
   c. Oxidation
   d. Reduction

6. Which complex does not pump protons?

   a. NADH dehydrogenase complex I
   b. Succinate dehydrogenase complex II
   c. cytochrome reductase complex III
   d. cytochrome oxidase complex IV

7. An uncoupler of oxidative phosphorylation is

   a. DNP
   b. DTT
   c. azide
   d. Rotenone

II Give short answer for the following

1. Explain Redox reactions.
2. Outline the structure of mitochondria.
3. Describe the inhibitors of electron transport chain.
4. What do you understand by oxidative phosphorylation?

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